LEVEL UP ACADEMY

Class 8 | CBSE / GSEB | Science

Quick Revision: Microorganisms: Friend and Foe

The main groups

- Microorganisms are too small to see without a microscope. Main groups: **bacteria, fungi, protozoa, algae**. **Viruses** are different: they reproduce only inside the cells of a host.
- They live everywhere: in soil, water, air, and inside our bodies.

Friends (useful)	How
Curd	The bacterium Lactobacillus turns milk into curd
Bread, cakes	Yeast respires and gives off carbon dioxide, which makes the dough rise
Alcohol, wine	Yeast turns sugar into alcohol (fermentation)
Medicines	Antibiotics such as penicillin (discovered by Alexander Fleming) are made from bacteria and fungi
Vaccines	Dead or weakened microbes train the body to fight a disease (Edward Jenner: smallpox)
Soil fertility	Rhizobium in the root nodules of legumes fixes nitrogen; decomposers recycle dead matter

Disease	Caused by	Spread by
Cholera, typhoid	Bacteria	Contaminated water and food
Tuberculosis	Bacteria	Air (coughing, sneezing)
Malaria	Protozoan	Female Anopheles mosquito
Dengue	Virus	Aedes mosquito
Measles, chickenpox, polio	Virus	Air / contact / water
In plants: citrus canker; rust of wheat; yellow vein mosaic of okra	Bacteria; fungi; virus	Air, insects

Food preservation: salt (pickles, fish), sugar (jams), oil and vinegar (pickles), heating and cooling, airtight packing.
Pasteurisation (Louis Pasteur): milk is heated to about 70 °C for 15 to 30 seconds, then quickly chilled.

Quick check

1. Name the microorganism that makes dough rise, and explain how it does it.

2. How do the root nodules of pea and gram plants help the soil?